

# Soumitri Chattopadhyay

UC San Diego, CA, USA

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[Google Scholar](#) | [GitHub](#) | [LinkedIn](#)

## RESEARCH INTERESTS

**Methods:** Foundational Models, Multi-modal learning (Vision+X), In-Context learning, Visual Agentic AI

**Applications:** Medical Image Computing, Longitudinal Analysis, Clinical Workflows; Image Restoration

## EDUCATION

**University of California, San Diego**

*Ph.D. in Computer Science and Engineering*

California, USA

*Sept. 2025 – May 2028 (Expected)*

**University of North Carolina at Chapel Hill**

*M.S. in Computer Science (Ph.D. Transferred to UCSD)*

North Carolina, USA

*Aug. 2023 – May 2025*

**Jadavpur University**

*B.E. in Information Technology | CGPA (out of 10): 9.01*

Kolkata, India

*Aug. 2019 – June 2023*

## RESEARCH EXPERIENCES

**UC San Diego**

*Graduate Student Researcher*

California, USA

*Sept. 2025 – Present*

- Part of the [UCSD Biomedical Image Analysis Group](#), advised by [Prof. Marc Niethammer](#).
- Currently working on in-context promptable foundational models for multimodal 2D/3D medical segmentation.
- Devised cross-modal visual prompting approaches for 3D medical segmentation with foundational models.
- Analyzed robustness of segmentation foundational models against imprecise visual prompts (**ISBI ‘26 Oral**).
- Developed a visual agentic workflow to process, segment and extract radiomics features from 2D and 3D raw clinical imaging data for early diagnosis of Lupus (as a part of an **NIH grant**).

**UNC Chapel Hill**

*Graduate Research Assistant*

North Carolina, USA

*Aug. 2023 – Sept. 2025*

- Part of the erstwhile [UNC-BIAG](#), advised by [Prof. Marc Niethammer](#). Transferred to UCSD upon advisor’s move.
- Conducted an extensive study on zero-shot domain generalizability of medical segmentation foundational models.
- Proposed a few-shot latent diffusion model-aided fine-grained image recognition (**BMVC ‘24 Oral**).

**SRI International**

*Research Intern*

Menlo Park, CA

*May 2024 – Dec. 2024*

- Interned in the [NuSCI Research Group](#), supervised by [Dr. Anirban Roy](#)
- Worked on **brain decoding**; developed an end-to-end framework for visual decoding from fMRI scans with controllable latent diffusion models and various training recipes that outperformed prior SOTA works.

**École de technologie supérieure (ÉTS), Montreal**

*MITACS Globalink Research Intern*

Montreal, Canada

*June 2022 – Aug. 2022*

- Advisor: [Prof. Jose Dolz](#)
- Explored handling of 3D medical volumes (CT and MRI), including pre-processing, sampling, and visualization.
- Implemented a knowledge distillation-aided baseline for unpaired multi-modal medical image segmentation.

**CVPR Unit, Indian Statistical Institute**

*Research Intern*

Kolkata, India

*Nov. 2021 – May 2023*

- Supervisors: [Prof. Umapada Pal](#) and [Prof. Saumik Bhattacharya](#)
- Investigated the role of temperature in Self-supervised Contrastive loss and proposed a dynamic temperature scaling to improve its performance (**IEEE TAI ‘24**).
- Developed Self-Supervised Learning frameworks for Offline Signature Verification (**ICIP ‘22 Oral, ICPR ‘22**).

## PREPRINTS

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2. D. Mandal, **S. Chattopadhyay**, Y. Wang, M. Niethammer, P. Chakravarthula, [UNSCAR: Universal, Scalable, Controllable, and Adaptable Image Restoration](#), *Under Review*.
1. **S. Chattopadhyay**, Y. Wang, D. Mandal, B. Demir, M. Niethammer, P. Chakravarthula, [Diffractive Optical Neural Networks for Ultra-Light-Weight Medical Image Segmentation](#), *Under Review*.

## SELECTED PUBLICATIONS

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13. **S. Chattopadhyay**, B. Demir, M. Niethammer, [Uncertainty-Aware Spatio-Semantic Contextual Prompts for Multimodal Medical Segmentation](#), **MICCAI 2026** (*Early Accepted, top-9%*)
12. B. Demir, **S. Chattopadhyay**, H. Greer, B. Chen, M. Niethammer, [How Useful Are Vision Foundation Model Features for Out-of-the-Box Disease Progression Prediction?](#), **CVPR 2026 – CV4Clinical Workshop**, Oral
11. **S. Chattopadhyay**, B. Demir, M. Niethammer, [On the Robustness of Foundational 3D Medical Image Segmentation Models Against Imprecise Visual Prompts](#), **ISBI 2026**, Oral
10. D. Mandal, **S. Chattopadhyay**, G. Tong, P. Chakravarthula, [UNICORN: Latent Diffusion-based Unified Controllable Image Restoration Network across Multiple Degradations](#), **WACV 2026**, Oral
9. **S. Chattopadhyay**, S. Biswas, E. Vivoli, J. Lladós, [Towards Generative Class Prompt Learning for Fine-grained Visual Recognition](#), **BMVC 2024**, Oral
8. A. Sain, A.K. Bhunia, S. Koley, P.N. Chowdhury, **S. Chattopadhyay**, T. Xiang, Y.Z. Song, [Exploiting Unlabelled Photos for Stronger Fine-Grained SBIR](#), **CVPR 2023**
7. H. Thakur and **S. Chattopadhyay**, [Active Learning for Fine-Grained Sketch-Based Image Retrieval](#), **BMVC 2023**
6. M. Bhattacharyya, **S. Chattopadhyay**, S. Nag, [DeCAtt: Efficient Vision Transformers with Decorrelated Attention Heads](#), **CVPR 2023 – ECV Workshop**, Oral
5. **S. Chattopadhyay**, S. Ganguly, S. Chaudhury, S. Nag, S. Chattopadhyay, [Exploring Self-Supervised Representation Learning For Low-Resource Medical Image Analysis](#), **ICIP 2023**
4. A. Deb, S. Nag, A. Mahapatra, **S. Chattopadhyay**, A. Marik, *et al.* [BEATS: Bengali Speech Acts Recognition using Multimodal Attention Fusion](#), **INTERSPEECH 2023**, Oral
3. H. Basak, **S. Chattopadhyay**, R. Kundu, S. Nag, R. Mallipeddi, [IDEAL: Improved DENSE local Contrastive Learning for Semi-Supervised Medical Image Segmentation](#), **ICASSP 2023**
2. S. Manna, **S. Chattopadhyay**, S. Bhattacharya, U. Pal, [SWIS: Self-Supervised Representation Learning For Writer Independent Offline Signature Verification](#), **ICIP 2022**, Oral
1. **S. Chattopadhyay**, S. Manna, S. Bhattacharya, U. Pal, [SURDS: Self-Supervised Attention-guided Reconstruction and Dual Triplet Loss for Offline Signature Verification](#), **ICPR 2022**

## TECHNICAL SKILLS

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**Frameworks/Libraries:** PyTorch, TensorFlow, MONAI, PyRadiomics, NumPy, Pandas, Matplotlib

**Softwares/Developer Tools:** LaTeX, VSCode, Slicer, ITK-SNAP, Android Studio

## ACADEMIC SERVICES

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### Peer-reviewing (Regular, since 2022)

- **Conferences:** CVPR, MICCAI, WACV, ISBI, ICCV, ACM MM, ICPR; ICCV + CVPR Workshops
- **Journals:** International Journal of Intelligent Systems, Wiley ; Engineering Applications of Artificial Intelligence, Elsevier ; Computers in Biology and Medicine, Elsevier ; IEEE Access